

Power Series for Solving ODEs

Ordinary Differential Equations

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Power Series

- As noted a few times, **not all differential equations have exact solutions**.
- So, we need to resort to seeking approximate solutions, or solutions in the neighborhood of the initial value. Before describing these methods, we need to recall the power series. A power series expansion about $x = x_0$ with coefficient sequence a_n is given by $\sum_{n=0}^{\infty} a_n (x - x_0)^n$.
- For now, we will consider all constants to be real numbers with x in some subset of the set of real numbers.

Power Series

$$\sum_{n=0}^{\infty} n! x^n, \quad \sum_{n=0}^{\infty} \frac{2^n}{n!} x^n$$

- We review power series briefly

$$\sum (-1)^n \frac{1}{(2n+1)!} x^n, \quad a_n = \frac{(-1)^n}{(2n+1)!}$$

Definition (Power Series)

An infinite series of the form

$$\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \dots \quad (1)$$

at $x=0$

is called a power series in x . The series

$$\sum_{n=0}^{\infty} a_n (x - x_0)^n = a_0 + a_1 (x - x_0) + a_2 (x - x_0)^2 + \dots \quad (2)$$

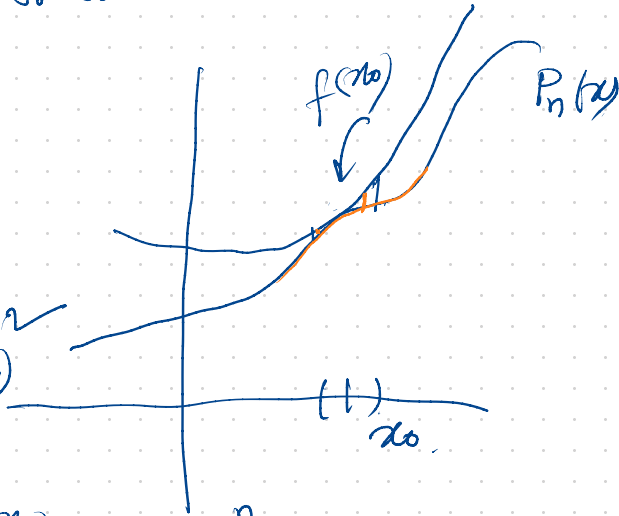
$(x_0 = 1) \rightarrow \sum_{n=0}^{\infty} \frac{1}{n!} (x-1)^n$

is a power series in $x - x_0$, and is somewhat more general than (1).

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}; \quad a_n = \frac{1}{n!}$$

$f(x), f(x_0), f'(x_0), f''(x_0), f'''(x_0), \dots, f^{(n)}(x_0)$
 $\underbrace{\hspace{10em}}_{n\text{-th order derivative } f(x)}$

$$f(x) - P_n(x) = R_n(x, x_0)$$



$$\begin{aligned}
 P_n(x) = & f(x_0) + \frac{f'(x_0)}{1!} (x-x_0) + \frac{f''(x_0)}{2!} (x-x_0)^2 \\
 & + \dots + \frac{f^{(n)}(x_0)}{n!} (x-x_0)^n
 \end{aligned}$$

$$R_n(x, x_0) = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x-x_0)^{n+1}$$

$$P_n(x) = \sum_{k=1}^{(n)} a_k (x-x_0)^k$$

$$a_k = \frac{f^{(k)}(x_0)}{k!}$$

$$\lim_{n \rightarrow \infty} P_n(x) \longrightarrow \sum_{k=1}^{\infty} a_k (x-x_0)^k$$



Power Series: Taylor Series

Remark

However, (2) can always be converted into (1) by substituting x for $x - x_0$, which simply corresponds to a shift of the coordinate system. Thus, in general, we will restrict our attention to power series of the form (1).

- The two types of series encountered in calculus are *Taylor* and *Maclaurin* series.

Taylor Series

A Taylor series expansion of $f(x)$ about $x = x_0$ is the Taylor series

$$f(x) \approx \sum_{n=0}^{\infty} a_n (x - x_0)^n, \quad (3)$$

where

$$a_n = \frac{f^{(n)}(x_0)}{n!} \quad (4)$$

Convergence of Power Series

Remark

Note that we use $\approx$ to indicate that we have yet to determine when the series may converge to the given function.

Definition (Convergence of a Power Series)

The series (1) is said to converge at a point x if the limit

$$\sum_{n=0}^{\infty} a_n x^n$$

$$\lim_{m \rightarrow \infty} \sum_{n=0}^m a_n x^n$$

$$\sum_{n=0}^{\infty} \frac{x^n}{n!} = \sum_{n=0}^{\infty} \frac{e^n}{n!} + \dots$$

exists, and in this case, the sum of the series is the value of this limit.

- It is obvious that (1) always converges at the point $x = 0$. With respect to the arrangement of their points of convergence, all power series in x fall into one or another or three major categories

Radius of Convergence

Theorem (Convergence of Power Series)

For any power series,

$$\sum_{n=0}^{\infty} a_n (x - x_0)^n,$$

exactly one of these statements is true:

- The power series converges only for $x = x_0$.
- The power series converges for all values of x .
- There's a positive number R such that the power series converges if $|x - x_0| < R$ and diverges if $|x - x_0| > R$.

→ $|x - x_0| = R \rightarrow$ no comments



Definition (Radius of Convergence)

In the last case, we say that R is the radius of convergence of the power series

Finding Radius of Convergence

Theorem (Test of Convergence)

Suppose there's an integer N such that $a_n \neq 0$ if $n \geq N$ and

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = L,$$

where $0 \leq L \leq \infty$. Then the radius of convergence of $\sum_{n=0}^{\infty} a_n(x - x_0)^n$, is $R = \frac{1}{L}$ which should be interpreted to mean that $R = 0$ if $L = \infty$ or $R = \infty$ if $L = 0$.

Example

Find the radius of convergence of the following series:

$a_n = n!$
 $a_{n+1} = (n+1)!$
 $\frac{a_{n+1}}{a_n} \rightarrow \infty$
 $R = \frac{1}{\infty} = 0$

$$\sum_{n=0}^{\infty} n! x^n, \quad \sum_{n=0}^{\infty} (-1)^n \frac{x^n}{n!}, \quad \sum_{n=0}^{\infty} 3^n n^2 (x-1)^n.$$

$a_n = \frac{1}{n!}$
 $a_n = 3^n n^2$

Term-by-Term Differentiation

Theorem (Term-by-Term Differentiation)

Suppose that the power series

$$f(x) = \sum_{n=0}^{\infty} a_n(x - x_0)^n \text{ for } |x - x_0| < R$$

has radius of convergence $R > 0$ and sum f . Then f is differentiable in $|x - x_0| < R$ and

$$f'(x) = \sum_{n=1}^{\infty} n a_n(x - x_0)^{n-1} \text{ for } |x - x_0| < R.$$

interchange of limit's order

Multiplication of Power Series

Definition (Multiplication of Two Power Series)

If $\sum_{n=0}^{\infty} a_n(x - x_0)^n$ has a positive radius of convergence ρ_1 and

$\sum_{n=0}^{\infty} b_n(x - x_0)^n$ has a positive radius of convergence ρ_2 , then term-wise

multiplication

$$\left(\sum_{n=0}^{\infty} a_n (x - x_0)^n \right) \left(\sum_{n=0}^{\infty} b_n (x - x_0)^n \right) =$$
$$(a_0 + a_1(x - x_0) + \cdots)(b_0 + b_1(x - x_0) + \cdots) = a_0 b_0 + [a_0 b_1 + a_1 b_0](x - x_0) + \cdots$$

is convergent for all

$$= \sum_{n=0}^{\infty} c_n (x - x_0)^n$$

$$|x - x_0| < \rho \text{ where } \rho = \min\{\rho_1, \rho_2\}.$$

$$c_n = \sum_{k=0}^n a_k b_{n-k}$$

Analytic Functions

Definition (Analytic Function)

A function f is said to be **analytic at** x_0 if, in an open interval about x_0 , the function is the sum of a power series

$$\sum_{n=0}^{\infty} a_n (x - x_0)^n$$

that has positive radius of convergence.

Remark

It turns out that this power series is exactly the Taylor expansion of f .

Shifting Index

Shifting the Summation Index in a Power Series

For any integer, the power series

The diagram illustrates the index shift process. On the left, two handwritten boxes show the substitution: $n+k = n_0$ and $n = n_0 - k$. In the center is the original series $\sum_{n=n_0}^{\infty} a_n (x - x_0)^{n-k}$. A blue arrow points from the n in the exponent to $n+k$ above it. Another blue arrow points from the n in the subscript to $n+k$ below it. A third blue arrow points from the n in the coefficient a_n to a_{n+k} below it.

$$\sum_{n=n_0}^{\infty} a_n (x - x_0)^{n-k}$$

can be rewritten as

$$\sum_{n=n_0-k}^{\infty} a_{n+k} (x - x_0)^n.$$

That is, replacing n by $n + k$ in the general term and subtracting k from the lower limit of summation leaves the series unchanged.

Introduction to Power Series Method

Historical Note

In the 18th century, it was commonly assumed that all “reasonable” functions (for instance, continuous ones) were analytic. This conviction strongly motivated the development of techniques based on power series. In the 19th century, however, it became clear that this assumption was false; most functions cannot be represented by any nice power series. Even so, the power series method has endured and remains a highly effective tool for solving ODEs today.

The Method in General

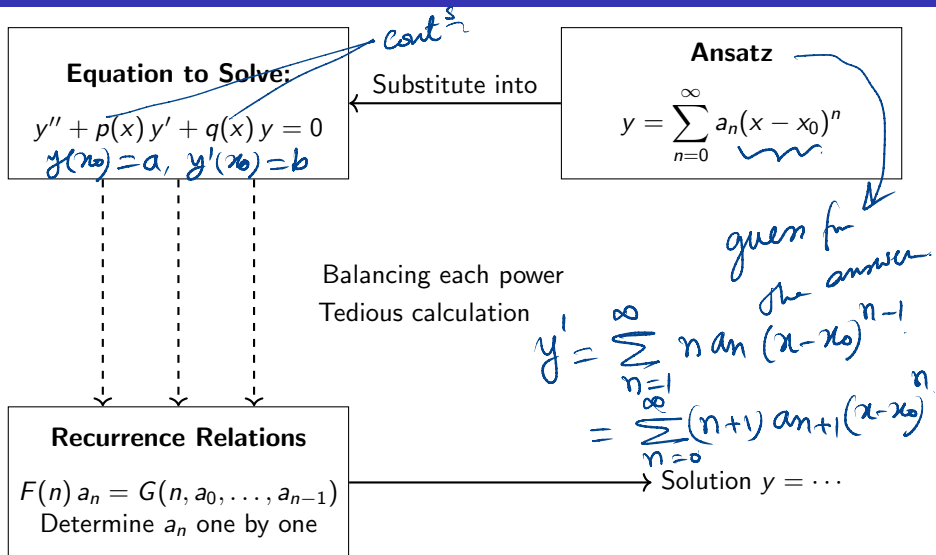


Figure: Naïve Power Series Method

Example -

$$y' - 2y = 0 \quad \text{with } y(0) = 1 \quad \boxed{x_0 = 0}$$

$$y = \sum_{n=0}^{\infty} a_n (x - x_0)^n = \sum_{n=0}^{\infty} a_n x^n \quad y = e^{2x} \cdot a$$

$\boxed{(x_0 = 0)} \rightarrow \text{Ansatz}$

$$y' = \sum_{n=0}^{\infty} (n+1) a_{n+1} x^n \quad (x_0 = 0)$$

Now substitute these identities into the eqnⁿ:

$$\sum_{n=0}^{\infty} (n+1) a_{n+1} x^n - 2 \sum_{n=0}^{\infty} a_n x^n = 0 = \sum_{n=0}^{\infty} 0 \cdot x^n$$

$$\boxed{(n+1) a_{n+1} - 2a_n = 0} \quad \forall n$$

$$n=0; \quad 1 \cdot a_1 - 2a_0 = 0 \Rightarrow a_1 = 2a_0 \Rightarrow a_{n+1} = \frac{2}{n+1} a_n$$

$$n=1; \quad 2a_2 - 2a_1 = 0 \Rightarrow a_2 = a_1 = 2a_0 \Rightarrow a_3 = \frac{2}{3} a_2 = \frac{2}{3} \times 2a_0$$

$$y(x) = \sum_{n=0}^{\infty} a_n x^n$$

$$= a_0 + 2a_0 x + 2a_0 x^2 + \frac{4}{3} a_0 x^3 + \dots$$

$$= a_0 \left(1 + 2x + 2x^2 + \frac{4}{3} x^3 + \dots \right) \quad \text{--- (Ans)}$$

$$= a_0 \left(1 + 2x + \frac{(2x)^2}{2!} + \frac{(2x)^3}{3!} + \dots \right)$$

$$y(0) = 1 = a_0 \left(1 + 2 \cdot 0 + \frac{(2 \cdot 0)^2}{2!} + \frac{(2 \cdot 0)^3}{3!} + \dots \right) a_0$$

$$= a_0 (1 + 0) = a_0 \Rightarrow a_0 = 1$$

Hence

$$y(x) = 1 + 2x + 2x^2 + \frac{4}{3} x^3 + \dots$$

Ans.

$$= \boxed{e^{2x}}$$

● Determine a power series Sol^n to the following linear IVP:
 $y' = (x-1)^2 y$ with $y(1) = -1$.

Solⁿ: Since IVP is given with $x_0 = 1$, our ansatz is $y(x) = \sum_{n=0}^{\infty} a_n (x-1)^n$.

$$\text{Now, } y' = \sum_{n=1}^{\infty} a_n n (x-1)^{n-1} \stackrel{\boxed{n \rightarrow n+1}}{=} \sum_{n=0}^{\infty} a_{n+1} (n+1) (x-1)^n.$$

$$\text{Now } y \text{ and } y' \rightarrow y' - (x-1)^2 y = 0.$$

$$\Rightarrow \sum_{n=0}^{\infty} (n+1) a_{n+1} (x-1)^n - (x-1)^2 \sum_{n=0}^{\infty} a_n (x-1)^n = 0.$$

$$\Rightarrow \sum_{n=0}^{\infty} (n+1) a_{n+1} (x-1)^n - \sum_{n=0}^{\infty} a_n (x-1)^{n+2} = 0$$

$$\Rightarrow a_1 + 2a_2(x-1) + \sum_{n=2}^{\infty} (n+1)a_{n+1}(x-1)^n - \sum_{n=0}^{\infty} a_n(x-1)^{n+2} = 0$$

$$\xrightarrow{n \rightarrow n+2} a_1 + 2a_2(x-1) + \sum (n+2+1)a_{n+2+1}(x-1)^{n+2} - \sum_{n=0}^{\infty} a_n(x-1)^{n+2} = 0$$

$$\Rightarrow a_1 + 2a_2(x-1) + \sum_{n=0}^{\infty} (n+3)a_{n+3}(x-1)^{n+2} - \sum_{n=0}^{\infty} a_n(x-1)^{n+2} = 0$$

$$\Rightarrow a_1 + 2a_2(x-1) + \sum_{n=0}^{\infty} ((n+3)a_{n+3} - a_n)(x-1)^{n+2} = \sum_{n=0}^{\infty} 0 \cdot (x-1)^n$$

Now we compare the same exponents, to get

$$a_1 = 0, \quad 2a_2 = 0 \Rightarrow a_2 = 0$$

and

$$(n+3)a_{n+3} - a_n = 0, \quad \forall n \geq 0$$

$$= 0 + 0 \cdot (x-1) + 0 \cdot (x-1)^2 + \dots$$

(Imp)

$$\Rightarrow a_{n+3} = \frac{1}{n+3} a_n \quad \forall n \geq 0$$

therefore $a_3 = \frac{1}{3} a_0$ ($n=0$)

$$a_4 = \frac{1}{4} \cdot a_1 = 0; \quad a_5 = \frac{1}{5} \cdot a_2 = 0.$$

($n=1$) ($n=2$)

$$a_6 = \frac{1}{6} a_3 = \frac{1}{6} \cdot \frac{1}{3} a_0 = \frac{1}{6 \cdot 3} a_0.$$

$$a_7 = 0 = a_8, \quad a_9 = \frac{1}{9} a_6 = \frac{1}{9 \cdot 6 \cdot 3} a_0.$$

only a_{3n} terms will be survived!

$$a_{3n} = \frac{1}{3^n n!} a_0$$

$$\forall n \geq 1. \quad \int \frac{1}{x} dx$$

Final general solⁿ

$$y(x) = a_0 + a_3(x-1)^3 + a_6(x-1)^6 + \dots$$

$$= \sum_{n=0}^{\infty} a_{3n} (x-1)^{3n}$$

where

$$a_{3n} = \frac{a_0}{3^n n!}$$

$$a_0 = -1$$

$$y(x) = \sum_{n=0}^{\infty} \frac{(-1)}{3^n n!} (x-1)^{3n} \quad (\text{Ans})$$

● Power Series Solⁿ for 2nd order homogeneous ODE

We now discuss the power series solⁿ for 2nd order IVPs:

$$(N) \text{ — } y'' + p(x)y' + q(x)y = g(x), \quad y(x_0) = k_0, \quad y'(x_0) = k_1$$

where $p(x)$, $q(x)$ and $g(x)$ are analytic at x_0 .

If $g(x) = 0$ then for the homogeneous case,

$$y'' + p(x)y' + q(x)y = 0 \text{ — (H)}$$

We say x_0 to be an **ORDINARY POINT** if $p(x)$ and $q(x)$ are analytic at x_0

Polynomials are analytic.

Let's say $P_1(x) = x - 2$

Qn- Is $P_1(x)$ analytic at $x=0$?

Ans. YES

$$P_1(x) = x - 2 = \sum_{n=0}^{\infty} a_n x^n$$

$$a_0 = -2, a_1 = 1, a_2 = 0, a_3 = 0, \dots, a_n = 0 \text{ for } n \geq 2$$

Qn. Is $P_1(x)$ analytic at $x=2$? Ans YES

$$P_1(x) = (x-2) = \sum_{n=0}^{\infty} a_n (x-2)^n, \quad a_0 = 0, a_1 = 1, a_n = 0, \forall n \geq 2,$$

Example

Example

Find at least the first four nonzero terms in a power series expansion about $x = 0$ for a general solution to

$$y'' - x^2 y = 0.$$

Ansatz, $y(x) = \sum_{n=0}^{\infty} a_n x^n$

$$\Rightarrow y'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1} = \sum_{n=0}^{\infty} (n+1) a_{n+1} x^n$$

$$\Rightarrow y''(x) = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} \stackrel{n \rightarrow n+2}{=} \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n$$

Now place the expressions of y and y'' in the given ODE.

$$y'' - x^2 y = 0.$$

$$\Rightarrow \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n - x \sum_{n=0}^{\infty} a_n x^n = 0.$$

$$\Rightarrow \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n - \sum_{n=0}^{\infty} a_n x^{n+2} = 0.$$

$$\Rightarrow 2a_2 + 6a_3 x + \sum_{n=2}^{\infty} (n+2)(n+1) a_{n+2} x^n - \sum_{n=2}^{\infty} a_{n-2} x^n = 0.$$

$$\Rightarrow 2a_2 + 6a_3 x + \sum_{n=2}^{\infty} ((n+2)(n+1) a_{n+2} - a_{n-2}) x^n = 0$$

Thus, we have

$$2a_2 = 0, \quad 6a_3 = 0 \quad \text{and} \quad a_{n+2} = \frac{1}{(n+2)(n+1)} a_{n-2}, \quad \boxed{n \geq 2}$$

That is,

$$a_2 = 0, a_3 = 0, a_4 = \frac{1}{4 \cdot 3} a_0 = \frac{1}{12} a_0, a_5 = \frac{1}{5 \cdot 4} a_1 = \frac{1}{20} a_1$$

Hence first four terms are

$$y(x) = a_0 + a_1 x + \frac{a_0}{12} x^4 + \frac{a_1}{20} x^5 + \dots$$

Example

Example

Find a power series expansion about $x = 0$ for a general solution to the given differential equation:

$$y'' - xy' + 4y = 0.$$

Your answer should include a general formula for the coefficients.

EXERCISE

- Example: Solve the following IVP:
 $y'' + 2y' - xy = 1 + 4x$; $y(0) = 0$ and $y'(0) = 0$

Comparing with $y'' + p(x)y' + q(x)y = g(x)$

$$p(x) = 2, q(x) = -x \text{ and } g(x) = 1 + 4x$$

all are analytic at $x_0 = 0$.

Our ansatz is: $y = \sum_{n=0}^{\infty} a_n x^n$

$$\text{Since } y' = \sum_{n=0}^{\infty} (n+1) a_{n+1} x^n$$

$$\text{and } y'' = \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n$$

Now put the expressions for y, y' and y'' we get

$$y'' + 2y' - xy = \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n + 2 \sum_{n=0}^{\infty} (n+1) a_{n+1} x^n - x \sum_{n=0}^{\infty} a_n x^n$$

$$= 2a_2 + \sum_{n=1}^{\infty} (n+2)(n+1) a_{n+2} x^n + 2a_1 + 2 \sum_{n=1}^{\infty} (n+1) a_{n+1} x^n - \sum_{n=0}^{\infty} a_n x^{n+1}$$

$$= 2a_2 + \sum_{n=0}^{\infty} (n+3)(n+2) a_{n+3} x^{n+1} + 2a_1 + 2 \sum_{n=0}^{\infty} (n+2) a_{n+2} x^{n+1} - \sum_{n=0}^{\infty} a_n x^{n+1}$$

$$= 2a_2 + 2a_1 + (6a_3 + 4a_2 - a_0) + \sum_{n=1}^{\infty} ((n+3)(n+2)a_{n+3} + 2(n+2)a_{n+2} - a_n) x^{n+1}$$

$$= 1 + 4x + 0x^2 + 0x^3 + \dots$$

By comparing,

$$2a_2 + 2a_1 = 1, \quad 6a_3 + 4a_2 - a_0 = 4$$

$$\text{and } (n+3)(n+2)a_{n+3} + 2a_{n+2}(n+2) - a_n = 0.$$

Hence $a_0 = 0$ and $a_1 = 0$, from the initial condⁿ.

$$a_2 = \frac{1-2a_1}{2} = \frac{1}{2},$$

$$a_3 = \frac{4 + a_0 - 4a_2}{6} = \frac{1}{3}$$

$$a_{n+3} = \frac{2(n+2)a_{n+2} - a_n}{(n+3)(n+2)} \quad \forall n \geq 2$$

$$a_4 = -\frac{a_3}{2} = -\frac{1}{6}$$

$$a_5 = -\frac{2}{5}a_4 + \frac{1}{20}a_2$$

$$a_6 = -\frac{1}{3}a_5 + \frac{1}{20}a_3$$

By calculating first few terms we get

$$y = \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{6}x^4 + \frac{11}{120}x^5 - \frac{7}{360}x^6 + \dots$$

(Ans)

Thank You

Questions?